

Nonlinear Modeling, Control, and Estimation of an Underactuated Pendulum System

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Abstract

This report develops a first-principles dynamic model of a reaction-wheel-actuated pendulum using Lagrangian mechanics. A state-space representation is derived and used to design a Linear Quadratic Regulator (LQR) for stabilisation near the unstable upright equilibrium. A bang-bang energy-shaping controller handles the swing-up phase. Simulation results confirm stable regulation from the downward resting position.

1 Problem Statement

1.1 System Description

A rigid pendulum is mounted on a pivot and actuated by a reaction wheel attached at its tip. The wheel is driven by a motor that applies an internal torque τ between the pendulum body and the wheel. By Newton's third law, this torque acts equally and oppositely on both components. The system is planar and the only external force is gravity.

1.2 Objective

Develop a first-principles dynamic model using Lagrangian mechanics, derive a state-space representation suitable for control design, and implement a two-phase controller: (1) a bang-bang energy pump to swing the pendulum from rest to near the upright position, and (2) an LQR to stabilise the upright unstable equilibrium.

1.3 Why the Problem Is Nontrivial

Three sources of difficulty make this problem non-trivial:

- **Nonlinearity.** The governing equation contains $\sin \theta$, which is only linear near equilibrium. A linear controller alone cannot handle large-angle motion.
- **Underactuation.** The system has two degrees of freedom (θ and ϕ) but only one control input (τ). The wheel angle ϕ is not independently controllable.
- **Unstable equilibrium.** The upright position is open-loop unstable. Any perturbation causes the pendulum to fall without active control.

2 Physical Model

2.1 Convention

Throughout this report the following angle convention is used, consistent with the simulation:

Value	Physical meaning
$\theta = 0$	Pendulum hanging down (stable equilibrium)
$\theta = \pi$	Pendulum pointing up (unstable equilibrium, control target)

This is the natural physical convention: gravity acts to restore the pendulum toward $\theta = 0$.

2.2 Assumptions

- Massless and rigid pendulum rod
- No friction, no air resistance (ideal case; extensions in Section 6)
- Planar motion only
- Ideal motor: torque τ is applied instantaneously and without loss
- Gravitational acceleration g is constant

2.3 Generalised Coordinates

Symbol	Description	Convention
θ	Pendulum angle measured from downward vertical	$\theta = 0$ is down
ϕ	Reaction wheel angle relative to the pendulum body	

The absolute angular position of the reaction wheel in the inertial frame is $\theta + \phi$.

2.4 Parameters

Symbol	Description	Value used
m	Pendulum bob mass	0.5 kg
l	Pendulum length	0.3 m
I_w	Wheel moment of inertia	0.002 kg m ²
g	Gravitational acceleration	9.81 m/s ²

3 Equations of Motion

3.1 Kinetic Energy

The system has two contributions to kinetic energy.

Pendulum bob (point mass, translational):

$$T_{\text{pend}} = \frac{1}{2}ml^2\dot{\theta}^2$$

Reaction wheel (rotational about pivot):

$$T_{\text{wheel}} = \frac{1}{2} I_w (\dot{\theta} + \dot{\phi})^2$$

Total kinetic energy:

$$T = \frac{1}{2} m l^2 \dot{\theta}^2 + \frac{1}{2} I_w (\dot{\theta} + \dot{\phi})^2$$

3.2 Potential Energy

With $\theta = 0$ defined as the downward position, the bob is at its lowest point when $\theta = 0$ and at its highest when $\theta = \pi$. The potential energy is:

$$V = -mgl \cos \theta$$

Verification:

- $\theta = 0$ (down): $V = -mgl$ (minimum, stable) ✓
- $\theta = \pi$ (up): $V = +mgl$ (maximum, unstable) ✓

3.3 Lagrangian

$$\mathcal{L} = T - V = \frac{1}{2} m l^2 \dot{\theta}^2 + \frac{1}{2} I_w (\dot{\theta} + \dot{\phi})^2 + mgl \cos \theta$$

3.4 Euler–Lagrange Equations

Applying the Euler–Lagrange equation to each generalised coordinate q_i :

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i$$

The motor applies torque τ internally between the pendulum and the wheel. By Newton’s third law, the generalised forces are:

$$Q_\theta = -\tau, \quad Q_\phi = +\tau$$

For θ :

$$(m l^2 + I_w) \ddot{\theta} + I_w \ddot{\phi} - mgl \sin \theta = -\tau$$

For ϕ :

$$I_w (\ddot{\theta} + \ddot{\phi}) = \tau$$

Note the sign on $\sin \theta$: with $V = -mgl \cos \theta$, the partial derivative $\partial V / \partial \theta = mgl \sin \theta$ enters with a negative sign in the Euler–Lagrange equation, giving $-mgl \sin \theta$ on the left-hand side. This is consistent with a restoring force toward $\theta = 0$.

3.5 Solving for Accelerations

From the ϕ equation:

$$\ddot{\phi} = \frac{\tau}{I_w} - \ddot{\theta}$$

Substituting into the θ equation and simplifying:

$$\ddot{\theta} = -\frac{g}{l} \sin \theta - \frac{2}{ml^2} \tau$$

$$\ddot{\phi} = \frac{\tau}{I_w} + \frac{g}{l} \sin \theta + \frac{2}{ml^2} \tau$$

Physical check: with $\tau = 0$ (no control):

- Near $\theta = 0$ (down): $\sin \theta \approx \theta$, so $\ddot{\theta} \approx -(g/l)\theta < 0$ for $\theta > 0$. Restoring. Stable. ✓
- Near $\theta = \pi$ (up): let $\delta = \theta - \pi$, then $\sin \theta = \sin(\pi + \delta) \approx -\delta$, so $\ddot{\theta} \approx +(g/l)\delta$. Diverging. Unstable. ✓

4 State-Space Formulation

4.1 Reduction to Two States

Although the full system has three states $[\theta, \dot{\theta}, \dot{\phi}]$, note that $\dot{\phi}$ does not appear in the $\ddot{\theta}$ equation. The controllability matrix of the full 3×3 system has rank 2: the $\dot{\phi}$ direction is uncontrollable. Applying LQR to the full system produces a spurious gain on $\dot{\phi}$ that has no physical effect on the pendulum angle.

The correct approach is to design the LQR on the controllable subsystem:

$$x = \begin{bmatrix} \delta \\ \dot{\delta} \end{bmatrix}, \quad \delta = \theta - \pi$$

where δ is the deviation from the upright equilibrium, wrapped into $[-\pi, \pi]$.

4.2 Linearisation around Upright ($\theta = \pi$)

Near upright, $\sin(\pi + \delta) \approx -\delta$, so:

$$\ddot{\delta} \approx \frac{g}{l} \delta - \frac{2}{ml^2} \tau$$

The linearised state-space model is:

$$\dot{x} = Ax + B\tau, \quad A = \begin{bmatrix} 0 & 1 \\ \frac{g}{l} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ -\frac{2}{ml^2} \end{bmatrix}$$

Eigenvalues of A : $\lambda = \pm\sqrt{g/l} \approx \pm 5.72$ rad/s. The positive eigenvalue confirms open-loop instability at the upright equilibrium.

5 Control Design

5.1 Overview: Two-Phase Strategy

The controller switches between two modes based on proximity to the upright equilibrium:

1. **Swing-up** (large angles): bang-bang energy pump
2. **Stabilisation** (near upright): LQR

Switching condition: engage LQR when $|\delta| < 0.3$ rad and $|\dot{\delta}| < 2.0$ rad/s.

5.2 LQR Stabilisation

The LQR minimises the cost functional:

$$J = \int_0^\infty \left(x^\top Q x + R \tau^2 \right) dt$$

with weighting matrices:

$$Q = \begin{bmatrix} 300 & 0 \\ 0 & 10 \end{bmatrix}, \quad R = 1$$

The feedback law is $\tau = -Kx$ where K is the solution to the algebraic Riccati equation. The high weight on δ relative to $\dot{\delta}$ prioritises angle correction over velocity damping.

5.3 Bang-Bang Swing-Up

The LQR is only valid near the upright equilibrium. For large angles, a bang-bang energy pump is used:

$$\tau = \begin{cases} u_{\max} \cdot \text{sgn}(\dot{\theta} \cos(\theta - \pi)) & \text{if } E < E^* \\ 0 & \text{if } E \geq E^* \end{cases}$$

where the target energy is $E^* = mgl$ (upright with zero velocity) and the current energy is:

$$E = \frac{1}{2}ml^2\dot{\theta}^2 - mgl \cos \theta$$

The sign term $\dot{\theta} \cos(\theta - \pi)$ is positive when motion is directed toward the upright position, ensuring torque always adds energy rather than opposing it. When sufficient energy has been accumulated ($E \geq E^*$), torque is cut to zero and the pendulum coasts toward the top where LQR engages.

This formulation avoids two failure modes of the standard Åström energy-shaping law: (1) zero torque at startup due to the $\dot{\theta}$ factor, and (2) zero torque at the horizontal position where $\cos(\delta) = 0$.

6 Planned Extensions

6.1 Viscous Friction

Add a damping term to the equation of motion:

$$\ddot{\theta} = -\frac{g}{l} \sin \theta - \frac{b}{ml^2} \dot{\theta} - \frac{2}{ml^2} \tau$$

where b is a viscous damping coefficient. This reduces available energy per swing and tests whether the bang-bang controller has sufficient authority.

6.2 Motor Back-EMF

Replace the ideal torque input with a motor electrical model:

$$\tau = K_t \cdot I, \quad I = \frac{V_{\text{applied}} - K_e \dot{\phi}}{R_m}$$

where K_t is the torque constant, K_e the back-EMF constant, and R_m the winding resistance. This caps available torque at high wheel speeds and couples the electrical and mechanical dynamics.

6.3 State Estimation

Real sensors provide noisy measurements. An Extended Kalman Filter (EKF) or Luenberger observer will be added to estimate $[\theta, \dot{\theta}]$ from corrupted measurements before implementation on hardware.

6.4 Hardware Implementation

Target platform: STM32 F446RE Nucleo board (180 MHz Cortex-M4 with FPU). Sensing via IMU (MPU6050) with complementary filter for angle estimation. Control loop target: 500–1000 Hz via hardware timer interrupt.